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Ontology and Process Design Principles

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zoundra · Mar 2025

I was recently asked to describe how an "ideal" Ontology is designed. I don't know about you, such a thing as a "perfect" Ontology of it deserves a place for pride, if not, then it's bad. But, I thought I would share some ideas that will make your decision-making easier.

The Ontology

The Ontology is the APP of your organization, a shared layer between engineers, business users, and AI agents. It is composed of "intuitive business concepts" which allows us to communicate with each other.

The Ontology is designed to support operational decision-making. It provides the following capabilities:

- the relevant data for decisions
- the possible actions to record the decisions
- the logic to evaluate what decision to make

Ontology design is a critical skill that requires deliberate practice. The Ontology combines these nouns and verbs into unlimited sentences, allowing all of the types of logic and power decision making human reasoning, traditional business logic, linear optimizations, Generative AI, etc.

Design Approach

The Ontology is a shared resource. It's an APP that needs to be designed and maintained. Ontology design is a process that requires both business and tech. Use the Ontology Design Types and Actions and support versatile decision making.

- **Describe what the users do.** What decisions will they make? What information do they need to make those decisions? What actions do they take? What Ontology objects, and Actions, if the actions are available, the data can probably stay in Foundry.
- **Check the existing Ontology.** Once Object Types already exist, we need to only leverage and map already stored the connected.
- **Use Object Types to model the business concepts.** This will be a simple noun-based and the possibility exists to record the decisions. The Palantir Rubber Duck data model helper is very useful in this process.
- **Use Actions to record decisions.**
  - **Function Free** - Back-to-the-Ontology Objects, Actions, Applications using the Ontology SDK, Ontology Functions, Ontology Rules, Ontology Functions, Ontology Rules.
  - **Data Engineering Tools** - Integrate the data to fill out the Ontology backing datasets, Ontology Functions, Ontology Rules, Ontology Functions, Ontology Rules.
- **Build Integrated Workflows.** Check in with stakeholders regularly and business users regularly. Make sure that the logic is sound and the data exists.

Ontology design is a process and requires deliberate practice, and, if it's not perfect, it's a **foundation** for improvement, rather than, it's bad.

Design Rules

The purpose of the Ontology is to be shared and used company wide. Craft Object Types that you are proud to share and use.

Object Types

- **Object Types should have point of contact configured.**
- **Use a point of contact for Primary Keys** - Primary keys are critical for maintaining or depreciating the Object Type.
- **Use Shared Property Types** - Use the Ontology Manager Application.
- **Object Types should be up to date and trusting.**
  - Backing datasets should have schedules and health checks configured if they are critical to the business.
- **Use Object Type Statuses effectively.**
  - Experimental - Experimental Object Types are not intended for end-user consumption. Use this status for Object Types that are not ready for end-user consumption.
  - Active - The object type is stable and high quality. It adheres to the design guide and is ready for end-user consumption.
  - Deprecated - The object type has a production usage, redirected, low quality. It is not recommended for end-user consumption. Deprecated Object Types will be hidden.
- **Add the Object Type to the relevant project.**
  - Object Types are shared across Ontology Projects. Groups should be used instead of Ontologies.
- **Use Shared Property Types** of relevant nouns.
  - The Ontology should be the shared, trusted and better nouns, and verbs and give an extra opportunity for improvement. Select nouns that match similar, related concepts. Use the Ontology Manager application to find shared property types. Create new if not represent the same concept as the Object Type. Select.
- **Add relevant Aliases.**
  - Aliases are a powerful feature to be utilized to solve in different areas of the business, helping our Ontology. Use the Ontology Manager Application make relevant aliases for Object Types.
- **Fill out Object Type, Action, and Property descriptions.**

Primary and Foreign Keys

- **Pick a primary key** values **carefully and give thing. No exceptions.**
  - Use non-persistent numbers, not the other way around. Migrating primary key values is a pain. Use a UUID if there is no other option. Use the Ontology Manager and the Object Type is used, which is a high effort.
- **Use Strings for primary key values, Integers are fine, but no exceptions.**
  - The primary key must be constructed from properties of the Object Ontology only. If you don't have to change which column is the primary key. This potentially meaning you need to change which column is the primary key. This potentially meaning you need to change which column is the primary key. This potentially meaning you need